Valentino Cattaneo

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EDUCATION

Purdue University, **College of Engineering** | GPA: 3.94/4.00 — **West Lafayette, IN**
Bachelor of Science in Electrical Engineering — **May 2028**
Concentration: Electric Power and Energy Systems

PROFESSIONAL EXPERIENCE

Angel Aerial Systems — **Cincinnati, OH**
Electrical Engineering Intern — **May 2026 – August 2026**

- Led battery management system (BMS) root cause analysis on the Trio Scout UAV, identifying and fixing an SMBus timing flaw in open-source drivers
- Validated the fix across bench, thermal stress, and propulsion tests, restoring fleet operations within 2.5 weeks
- Engineered and validated BMS fault detection and mitigation algorithms, eliminating coverage gaps that allowed hardware faults to compromise a flight-critical system without detection or logging
- Redesigned the pouch-cell pack assembly (from soldering to spot-welding) to support UN 38.3 compliance
- Collaborated with connector vendors to select a 50A battery interface meeting airframe envelope constraints

Purdue Power Electronics and Drives Research Lab — **West Lafayette, IN**
Undergraduate Research Assistant — **June 2025 – December 2025**

- Developed a 6-phase Field-Oriented Control (FOC) motor drive prototype for a UAV propulsion system
- Designed and routed a magnetic encoder PCB in Altium, enabling 14-bit rotor position feedback over SPI
- Modeled two synchronized FOC control loops in Simulink, validating control design before hardware bring-up
- Tuned and validated current and speed PI gains on a propulsion test stand using oscilloscopes and LCR meters

TECHNICAL PROJECTS AND LEADERSHIP

Vertical Flight Systems Purdue, GoAERO VTOL Competition — **West Lafayette, IN**
President — **August 2025 – Present**

- Led a 30-member interdisciplinary team developing an autonomy-enabled 250 kg eVTOL
- Co-authored a conference paper at AIAA AVIATION 2026 on eVTOL development and flight testing
- Selected and integrated the avionics stack and the propulsion system for a 32kg PX4-based demonstrator
- Engineered the demonstrator's flight test campaign, from initial ground tests through fully autonomous flights
- Presented technical design reviews to faculty and industry advisors, summarizing risks, status, and action items

Vice President — **August 2024 – August 2025**

- Designed and routed a BMS for a 100 V, 14.4 kWh pack with SoC estimation, balancing, and protection systems
- Developed a MATLAB design-space optimizer for battery chemistries, rotor configurations, and propulsion systems using a convergent MTOW sizing loop to maximize aircraft payload
- Characterized Li-ion cells and sized battery modules to meet energy, discharge, and weight specifications
- Routed, soldered and validated all avionics and power circuits of the demonstrator platform
- Collaborated on ConOps, concept generation, and preliminary system design phases of the aircraft

Autonomous 3D-printed Indoor Multirotor - *Personal Project* — **May 2025 – August 2025**

- Programmed LiDAR SLAM, weighted A*, and frontier-based exploration in C++ on a Raspberry Pi companion computer with a PX4-integrated avionics stack, enabling autonomous, GNSS-denied indoors flight

TECHNICAL SKILLS

Design & Simulation: Altium, KiCAD, LTSpice, Simulink, Fusion360
Embedded: STM32 toolchain: CubeMX, CubeIDE & CubeProgrammer, Git
Programming: C++, C, Python, MATLAB